

HD-DVD went with the thicker protective layer so that, by virtue of its similarity to DVD, no hard coating was required; this also meant it was easier to make future drives backwards-compatible.

Indeed, backwards-compatibility with DVD is one of the things HD-DVD proponents frequently state. However, it might well turn out that Blu-ray drives will be forced to be backwards-compatible with DVD as well.

Early BDs were indeed susceptible to scratches, so much so that they had to be enclosed in plastic caddies. But in January of 2004, TDK introduced a proprietary clear polymer coating under the name 'Durabis'. This made BDs even more scratch resistant than DVDs! Now, BDs with the coating can also withstand attack by a screwdriver, according to a release on CNET News.com: "Researchers at TDK have developed a tough coating that will make scratched DVDs a thing of the past. In a test conducted by CNET News.com, a DVD treated with TDK's coating survived a determined attack with a screwdriver with no effect on playability."

Layers And Speeds

A single Blu-ray layer holds 25 GB; a single HD-DVD layer holds 15 GB. But it's not as simple as "Blu-ray is higher capacity than HD-DVD," as you'll find out.

In March of this year, at the Media Tech Expo 2005 in Las Vegas, the two camps demonstrated their current disc production capabilities. A standard two-layer HD-DVD-ROM is 30 GB. Toshiba showed off a three-layer HD-DVD-ROM, which is 45 GB; this was essentially their response to the criticism of HD-DVD's lower capacity. Now a dual-layer Blu-ray disc is 50 GB, but with 45 GB possible with three layers, HD-DVD had narrowed the difference to a mere 5 GB.

The capacity game itself is indicative of what's going on in the format war. For instance, in response to Toshiba, TDK, a member of the BDA, said it was working on a four-layer Blu-ray disc, which would have a capacity of 100 GB. And in June, it was reported that TDK had delivered on their promise—a prototype recordable Blu-ray disc

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with a capacity of 100 GB had been developed. It is, of course, the most advanced optical media developed to date.

What about speeds? Data transfer speeds, though not as important as capacity, are still an important factor in the adoption of any storage medium. Here, Blu-ray is ahead again by virtue of the specifications themselves—specifically, the NA and the wavelength. Blu-ray needs a lower disc rotation speed than HD-DVD to reach the same data transfer rates. 36 Mbps (1x) is the base speed for BDs, but BD-ROM movies will require a 54 Mbps data transfer rate—so the minimum speed the BDA is expecting to see in drives is 2x (72 Mbps).

Blu-ray also has the potential for much higher speeds. While the media itself limited recording speeds in the past, the only limiting factor for Blu-ray is the capacity of the hardware. The upper limit of today's drives in terms of constant rotation speed is about 10,000 rpm. This speed would result in a 12x (400 Mbps) transfer rate for a BD, and "only" 9x for an HD-DVD.

It's true that such high speeds are not required today, and HD-DVD proponents are likely to point out that 9x or 12x should make no difference to the end user—you. In reply, Blu-ray proponents would say that the 9x/12x difference showcase the superiority of their technology.

Where And When Can I Get Them?

We can provide a few indicators here: for the general consumer, 2006 seems to be the year. Players for Blu-ray and HD-DVD discs should arrive in stores in the US, Japan and many other countries sometime in 2006. BD-ROM pre-recorded media will probably be available by early 2006. Recorders should become available later in 2006 or in early 2007. It's not clear how much of a delay there will be in terms of availability in India.

Pioneer might soon introduce a Blu-ray writer for PCs. Toshiba might soon introduce drives for reading HD-DVDs in laptops. Imation, a member of the BDA and a name familiar in India, is finalising the development of both HD-DVD and Blu-ray media, utilising a protective disc coating for both formats. During 2006, Imation plans to launch BD-RE (rewritable) and BDR (recordable) 25 GB (single-layer) and 50 GB (dual-layer) media. At around the same time, the company also plans to launch the HD DVD-R (recordable) and HD DVD-ReR (Re-recordable) 15 GB single-layer and 30 GB dual-layer HD-DVD formats.

Who's On Whose Side?

There's news every week or so about someone having joined one camp or the other. Here are some of the more important or interesting facts.

The first consumer Blu-ray device in the US is expected to be PlayStation 3. In October, Paramount became the first studio to lend its support to both sides—earlier, it was part of the HD-DVD camp. Closer home, Moser Baer, interestingly, is on the HD-DVD side.

Prominent companies on the BDA (Blu-ray Disc Association) include Sony, TDK, Dell, Hitachi, Apple, HP, Philips and Samsung. The list of consumer electronics companies on Blu-ray's side is long—from Panasonic, Samsung, Sharp and Pioneer to LG Electronics and more.

